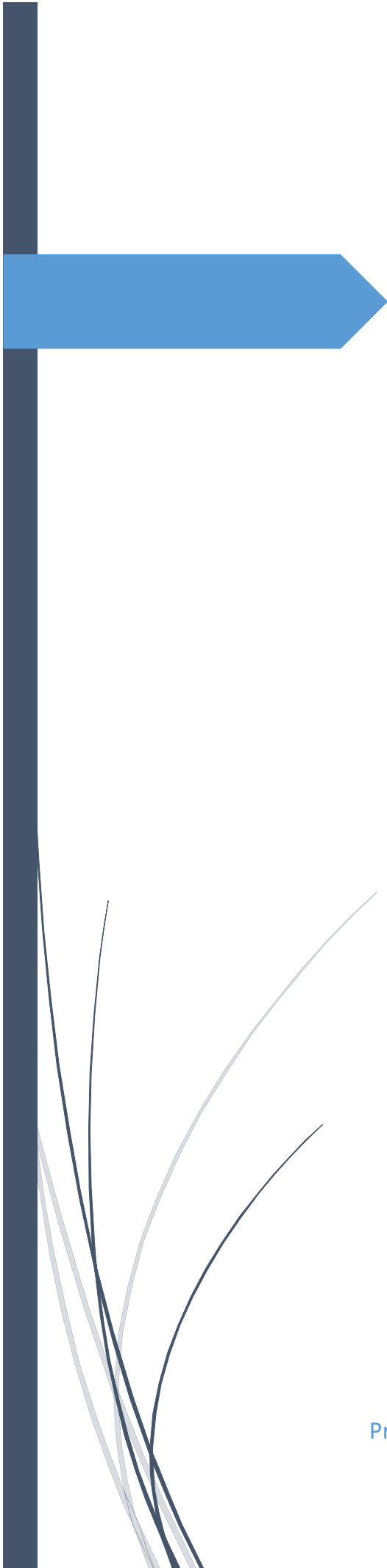




Linear Algebra

A Lecture note

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Syllabus

1. Algebra of Matrices:

Matrices, Transpose of a matrix, Square matrices, Powers of matrices, Polynomials in matrices, Invertible (Nonsingular) matrices, special types of square matrices, Complex matrices, and Block matrices.

2. Basic Operations:

Matrix addition and Scalar multiplication, Matrix multiplication, Calculation of Inverse, Row-echelon form, Elementary row and column operations, Rank.

Simultaneous Linear Equations: Augmented matrix, Gaussian elimination method, Pivoting strategies, Gauss-Jordan elimination.

3. Determinants:

Determinant of a square matrix, Minor, Cofactor, Inverse of a matrix using determinant, Pivotal condensation.

4. Vectors:

Dimension, Linear dependence and independence, Linear combinations.

Eigenvalues and Eigenvectors: Characteristic equations, Properties of eigenvalues and eigenvectors, linearly independent eigenvectors, Cayley-Hamilton theorem.

Book Reference:

- College Linear Algebra-Theory of Matrices ---- Prof. Md. Abdur Rahman
- Vector Analysis (Schum's outlines) – Murray R. Spiegel

1. Matrix Algebra

J. J. Sylvester was the first man who introduced the word ‘matrix’ in 1850 and later on in 1858 Arthur Cayley developed the theory of matrices in a systematic way.

A matrix is a rectangular array of numbers (real or complex) enclosed by a pair of brackets (or double vertical rolls) and the numbers in the array are called the entries or the elements of the matrix. That is, a rectangular array of (real or complex) numbers of the form

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

is called a matrix. The numbers $a_{11}, a_{12}, \dots, a_{mn}$ are called the entries or elements of the matrix. The above matrix has m rows and n columns and is called an $(m \times n)$ matrix. The matrix of m rows and n columns is said to be of order “ m by n ” or $m \times n$. The matrix can also be denoted by $[a_{ij}]$, $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$.

Hence a matrix can be defined as follow.

Definition of Matrix:

A *matrix* of size $m \times n$ over a field K , where m and n are positive integers, is an array with m rows and n columns, where each entry is an element of K . For $1 \leq i \leq m$ and $1 \leq j \leq n$, the entry in row i and column j of A is denoted by $[a_{ij}]$, and referred to as the (i, j) entry of A .

Example –

$A = \begin{bmatrix} 1 & 5 & 2 \\ -3 & 0 & 7 \end{bmatrix}$ is a matrix of order 2×3 over the real field $\mathbb{R}$ and also over the complex field $\mathbb{C}$.

Rows are $[1 \ 5 \ 2]$ and $[-3 \ 0 \ 7]$

and columns are $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$, & $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$

Addition and Scalar Multiplication of Matrices

Addition of the matrices is defined only for the matrices having same order. Let A and B be two matrices of order $m \times n$.

i.e.,

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \text{ and } B = [b_{ij}] = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{bmatrix}$$

Then the sum of A and B is

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

If $A = [a_{ij}]$ and $B = [b_{ij}]$, this takes the form

$$A + B = [a_{ij} + b_{ij}]$$

Example:

(1) If $A = \begin{bmatrix} 2 & 1 & 4 \\ -2 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -2 \\ 3 & 0 & 6 \end{bmatrix}$

Compute $A + B$.

Scalar Multiplication of Matrices –

The product of an $(m \times n)$ matrix A by a number k is denoted by kA or Ak is the $(m \times n)$ matrix obtained by multiplying every element of A by k , that is,

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & \dots & ka_{1n} \\ ka_{21} & ka_{22} & \dots & ka_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ka_{m1} & ka_{m2} & \dots & ka_{mn} \end{bmatrix}$$

Example –

(1) If $A = \begin{bmatrix} -2 & 0 \\ 3 & 4 \\ 1 & 5 \end{bmatrix}$, then compute $5A$.

Matrix Multiplication

Two matrices A and B are conformable for multiplication if the number of columns in A is equal to the number of rows in B .

Let $A = [a_1 \ a_2 \ \dots \ a_n]$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

Then $AB = [a_1b_1 \ a_2b_2 \ \dots \ a_nb_n] = [\sum_{i=1}^n a_ib_i]$

Again, let the $m \times p$ matrix $A = [a_{ij}]$ and the $p \times n$ matrix $B = [b_{ij}]$ then AB is the $m \times n$ matrix $C = [c_{ij}]$ where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj} = [\sum_{k=1}^p a_{ik}b_{kj}] ; i = 1, 2, \dots, m \text{ and } j = 1, 2, \dots, n$$

In the matrix product AB , the matrix A is called the pre-multiplier (or pre-factor), and B is called the post-multiplier (or post-factor)

Example-

If $A = \begin{bmatrix} 1 & -3 & 5 \\ 2 & 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -2 & 4 \\ 3 & 0 \end{bmatrix}$, Compute AB and BA and determine whether the matrices satisfy commutative law or not.

Special types of matrices:

Square Matrix:

If A is a $[a_{ij}]_{m \times n}$ matrix which have same number of rows and columns i.e. , $m = n$, then A is called a Square matrix.

Example: $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 5 & 4 \\ -3 & 2 & 1 \end{bmatrix}$ is a 3×3 square matrix.

Diagonal Matrix:

$A = [a_{ij}]_{m \times n}$, a square matrix is called a **diagonal matrix** in which the entries outside the main diagonal are all zero.

Example: $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$ is a 4×4 diagonal matrix.

Identity (Unit) Matrix:

A square matrix $A = [a_{ij}]_{m \times m}$ is said to be identity matrix if $a_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

Example: $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is a 4×4 identity matrix.

Transpose of a Matrix:

If A is an $m \times n$ matrix over the real field $\mathbb{R}$, then the $n \times m$ matrix obtained from the matrix A by writing its rows as columns and its columns as rows is called the transpose of A and is denoted by the symbol $A^T = [a_{ji}]$ is an $n \times m$ matrix.

For example, let $A = \begin{bmatrix} 1 & 0 & 5 & -7 \\ 2 & 3 & -1 & 6 \end{bmatrix}$, then $A^T = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 5 & -1 \\ -7 & 6 \end{bmatrix}$

Complex conjugate matrix:

The conjugate of a complex number $z = x + iy$ is the complex number $\bar{z} = x - iy$. If A is an $m \times n$ matrix over the complex field, then the conjugate of the matrix A is denoted by $\bar{A}$ whose elements are respectively the conjugate of the elements of A . i.e., if $A = [a_{ij}]$. Then $\bar{A} = [\bar{a}_{ij}]$.

For example,

if $A = \begin{bmatrix} 1 & i & 1+i \\ -i & 2 & 2+3i \\ -5 & 1+2i & -5 \end{bmatrix}$, then $\bar{A} = \begin{bmatrix} 1 & -i & 1-i \\ i & 2 & 2-3i \\ -5 & 1-2i & -5 \end{bmatrix}$

Real Matrix and Imaginary matrix:

A matrix A is called real provided it satisfies the relation $A = \bar{A}$.

Example: $\begin{bmatrix} 1 & 0 & 5 & -7 \\ 2 & 3 & -1 & 6 \end{bmatrix}$ is a 2×4 real matrix

A matrix A is called imaginary provided it satisfies the relation $A = -\bar{A}$.

Example: $\begin{bmatrix} 1 & -i & 1-i \\ i & 2 & 2-3i \\ -5 & 1-2i & -5 \end{bmatrix}$ is a 3×3 imaginary matrix

Conjugate transpose of a complex matrix:

The conjugate of the transpose of a given complex matrix A is said to be the conjugate transpose of A and generally denoted by the symbol A^* . That is

If $A = [a_{ij}]$ is a complex matrix, then

$$A^* = [\bar{A}^t] = [\bar{a}_{jl}]$$

For example, if $A = \begin{bmatrix} 1 & i & 1+i \\ -i & 2 & 2+3i \\ -5 & 1+2i & -5 \end{bmatrix}$, $A^* = \begin{bmatrix} 1 & i & 5 \\ -i & 2 & 1-2i \\ 1-i & 2-3i & -5 \end{bmatrix}$

Type of Matrix	Details
Row Matrix	$A = [a_{ij}]_{1 \times n}$
Column Matrix	$A = [a_{ij}]_{m \times 1}$
Zero or Null Matrix	$A = [a_{ij}]_{m \times n}$ where, $a_{ij} = 0$
Singleton Matrix	$A = [a_{ij}]_{m \times n}$ where, $m = n = 1$
Horizontal Matrix	$A = [a_{ij}]_{m \times n}$ where, $n > m$
Vertical Matrix	$A = [a_{ij}]_{m \times n}$ where, $m > n$
Square Matrix	$A = [a_{ij}]_{m \times n}$ where, $m = n$
Diagonal Matrix	$A = [a_{ij}]_{m \times n}$ where $a_{ij} = 0$ when $i \neq j$
Scalar Matrix	$A = [a_{ij}]_{m \times n}$ Where, $a_{ij} = \begin{cases} 0, & i = j \\ k, & i \neq j \end{cases}$ where k is a constant.
Identity (Unit) Matrix	$A = [a_{ij}]_{m \times m}$; Where, $a_{ij} = \begin{cases} 0, & i = j \\ 1, & i \neq j \end{cases}$

Type of Matrix	Details
Equal Matrix	$A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{r \times s}$ where, $a_{ij} = b_{ij}$, $m = r$, and $n = s$
Triangular Matrices	Can be either upper triangular ($a_{ij} = 0$, when $i > j$) or lower triangular ($a_{ij} = 0$ when $i < j$)
Singular Matrix	$ A = 0$
Non-Singular Matrix	$ A \neq 0$
Symmetric Matrices	$A = [a_{ij}]$ where, $a_{ij} = a_{ji}$
Skew-Symmetric Matrices	$A = [a_{ij}]$ where, $a_{ij} = -a_{ji}$
Hermitian Matrix	$A = A^*$
Skew – Hermitian Matrix	$A = -A^*$
Orthogonal Matrix	$A A^T = I_n = A^T A$
Idempotent Matrix	$A^2 = A$
Involutory Matrix	$A^2 = I, A^{-1} = A$
Nilpotent Matrix	$\exists p \in N$ such that $A^p = 0$
Normal Matrix	$AA^* = A^*A$
Unitary Matrix	$AA^* = A^*A = I$ or $A^* = A^{-1}$

Examples –

Singular matrix –

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ -1 & -1 & 7 \end{bmatrix}$$

Symmetric matrix–

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Skew-symmetric matrix –

$$A = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & -4 \\ -3 & -4 & 0 \end{bmatrix}$$

For any square matrix A, the symmetric part of the matrix is $-\frac{1}{2}(A + A^T)$

For any square matrix A, the skew - symmetric part of the matrix is $-\frac{1}{2}(A - A^T)$

Hermitian Matrix –

$A = \begin{bmatrix} 2 & 2-3i & 3 \\ 2+3i & 5 & 1+i \\ 3 & 1-i & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & i & -1-2i \\ -i & 3 & 5+3i \\ -1+2i & 5-3i & -1 \end{bmatrix}$ are Hermitian matrices.

Skew - Hermitian Matrix –

$$A = \begin{bmatrix} 2i & 2-3i & 3 \\ -2-3i & 5i & 1+i \\ -3 & -1+i & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} i & 1-i & 5 \\ -1-i & 2i & i \\ -5 & i & 0 \end{bmatrix}$$

Orthogonal Matrix –

$$A = \begin{bmatrix} \frac{1}{9} & \frac{8}{9} & -\frac{4}{9} \\ \frac{4}{9} & -\frac{4}{9} & -\frac{7}{9} \\ \frac{8}{9} & \frac{1}{9} & \frac{4}{9} \end{bmatrix} \text{ and } B = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix} \text{ are orthogonal matrices.}$$

Idempotent Matrix -

$$A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

Involutory matrix -

$$A = \begin{bmatrix} 4 & 3 \\ -5 & -4 \end{bmatrix} \text{ is an involutory matrix.}$$

Inverse Matrix:

A square matrix A is said to be invertible if there exists a unique matrix B such that $AB = BA = I$ where I is an identity matrix.

We call such a matrix B the inverse of A & is generally denoted by A^{-1} .

Example -

$$\text{Let } A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

$$\text{Then } AB = \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\text{and } BA = \begin{bmatrix} 4-3 & 2-2 \\ -6+6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Exercises -

$$(1) \text{ If } A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 5 & -1 \\ 2 & 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 5 & 3 \\ 7 & -2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \text{ then prove that } (AB)^T = B^T A^T$$

$$(2) \text{ Find the symmetric and skew-symmetric parts of the matrix } A = \begin{bmatrix} 1 & 2 & 4 \\ 6 & 8 & 1 \\ 3 & 5 & 7 \end{bmatrix}$$

$$(3) \text{ Prove that the matrix } A = \frac{1}{6} \begin{bmatrix} 3 & 3 & 3 & 3 \\ 3 & -5 & 1 & 1 \\ 3 & 1 & 1 & -5 \\ 3 & 1 & -5 & 1 \end{bmatrix} \text{ is orthogonal.}$$

$$(4) \text{ Show that the matrix } A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \text{ is involutory.}$$

Determination of inverse matrix –

Steps –

1. Singularity Check.

$$\text{If } D = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{vmatrix} \neq 0 \Rightarrow \text{Singular}$$

Proceed if non-singular. Stop otherwise. (means the matrix is invertible)

2. Finding Cofactors A_{ij} of the determinant of the matrix.

$$\text{For example, } A_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & \dots & a_{2n} \\ \vdots & \ddots & \vdots \\ a_{m2} & \dots & a_{mn} \end{vmatrix}$$

$$3. \text{ Construct the Adjoint/Adjugate matrix } Adj(A) = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}^T.$$

4. Thus $A^{-1} = \frac{1}{D} Adj(A)$ is the inverse.

Example –

$$(1) \text{ Find the inverse of the matrix } A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}.$$

Solution:

$$\text{Determinant of the matrix A is, } D = \begin{vmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{vmatrix}$$

$$= 3(-15) - 4(-4 - 6) - 1(5) = -45 + 40 - 5 = -10$$

So, A is a non-singular matrix and therefore, invertible.

Cofactors of A:

$$a_{11} = (-1)^{1+1} \begin{vmatrix} 0 & 3 \\ 5 & -4 \end{vmatrix} = -15$$

$$a_{12} = (-1)^{1+2} \begin{vmatrix} 1 & 3 \\ 2 & -4 \end{vmatrix} = 10$$

$$a_{13} = (-1)^{1+3} \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix} = 5$$

$$a_{21} = (-1)^{2+1} \begin{vmatrix} 4 & -1 \\ 5 & -4 \end{vmatrix} = 11$$

$$a_{22} = (-1)^{2+2} \begin{vmatrix} 3 & -1 \\ 2 & -4 \end{vmatrix} = -10$$

$$a_{22} = (-1)^{2+2} \begin{vmatrix} 3 & -1 \\ 2 & -4 \end{vmatrix} = -10$$

$$a_{23} = (-1)^{2+3} \begin{vmatrix} 3 & 4 \\ 2 & 5 \end{vmatrix} = -7$$

$$a_{31} = (-1)^{3+1} \begin{vmatrix} 4 & -1 \\ 0 & 3 \end{vmatrix} = 12$$

$$a_{32} = (-1)^{3+2} \begin{vmatrix} 3 & -1 \\ 1 & 3 \end{vmatrix} = -10$$

$$a_{33} = (-1)^{3+3} \begin{vmatrix} 3 & 4 \\ 1 & 0 \end{vmatrix} = -4$$

$$\text{So, Adj(A)} = \begin{bmatrix} -15 & 10 & 5 \\ 11 & -10 & -7 \\ 12 & -10 & 4 \end{bmatrix}^T = \begin{bmatrix} -15 & 11 & 12 \\ 10 & -10 & -10 \\ 5 & -7 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-10} \begin{bmatrix} -15 & 11 & 12 \\ 10 & -10 & -10 \\ 5 & -7 & -4 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & -\frac{11}{10} & -\frac{6}{5} \\ -1 & 1 & 1 \\ -\frac{1}{2} & \frac{7}{10} & \frac{2}{5} \end{bmatrix} \quad (\text{Ans.})$$

Solution of SLE using Matrix inverse -

3.

Solve the following linear equations with the help of inverse of matrices –

$$3x + 5y - 7z = 13$$

$$4x + y - 12z = 6 \quad \dots\dots\dots(1)$$

$$2x + 9y - 3z = 20$$

Solution:

The given linear equation can be written in matrix form as

$$\begin{bmatrix} 3 & 5 & -7 \\ 4 & 1 & -12 \\ 2 & 9 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 13 \\ 6 \\ 20 \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} 3 & 5 & -7 \\ 4 & 1 & -12 \\ 2 & 9 & -3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 13 \\ 6 \\ 20 \end{bmatrix}$$

Then the equation given by (1) reduces to $AX = I$ (2)

Let D be the determinant of matrix A.

$$\text{So, } D = \begin{vmatrix} 3 & 5 & -7 \\ 4 & 1 & -12 \\ 2 & 9 & -3 \end{vmatrix} = 3(-3 + 108) - 5(-12 + 24) - 7(36 - 2) = 315 - 60 - 238 = 17 \neq 0$$

Therefore, the matrix A is non-singular and hence, A^{-1} exists.

We multiply both sides of the equation no (2) by A^{-1} on the left. Then we get

$$A^{-1}AX = A^{-1}B \text{ or } X = A^{-1}B \text{ (3)}$$

Compute $A^{-1}B$ and determine the solution. (Left for students)

$$\text{Ans : } (A^{-1}B = X = \frac{1}{17} \begin{bmatrix} 17 \\ 34 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix})$$

Exercises –

$$(1) \text{ Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 1 & 3 & 2 & 5 \\ 2 & 1 & -1 & 3 \end{bmatrix}. \text{ Calculate the product AB.}$$

$$(2) \text{ If } A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \text{ Prove that } A^3 + A^2 - 21A - 54I = 0, \text{ Where I is the identity matrix of order } 3 \times 3.$$

$$(3) \text{ If } A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix} \text{ find } A^{-1}B.$$

$$(4) \text{ Find the inverse of matrix } A = \begin{bmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{bmatrix}.$$

(5) If $A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 5 & -1 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 & 3 \\ 7 & -2 & 1 \\ 2 & 0 & -3 \end{bmatrix}$ then prove that $(AB)^T = B^T A^T$.

(6) Find the symmetric and skew symmetric parts of the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 6 & 8 & 1 \\ 3 & 5 & 7 \end{bmatrix}$.

(7) Prove that the matrix $A = \frac{1}{6} \begin{bmatrix} 3 & 3 & 3 & 3 \\ 3 & -5 & 1 & 1 \\ 3 & 1 & 1 & -5 \\ 3 & 1 & -5 & 1 \end{bmatrix}$

(8) Show that the matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is involuntary.

(9) Find the inverse of the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{bmatrix}$.

(10) Find the inverse of the following matrix $A = \begin{bmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{bmatrix}$.

(11) Solve the following SLEs with the help of inverse of matrices –

(i)	(ii)	(iii)	(iv)
$x + y + z = 6$	$3x + 2y - z = 20$	$2x + y + z = 3$	$x + y + z = 3$
$x - y + z = 2$	$2x + 3y - 3z = 7$	$3x - 2y + 2z = 9$	$x + 2y + 2z = 4$
$2x + y - z = 1$	$x - y + 6z = 41$	$4x + 3y - z = -1$	$x + 4y + 9z = 6$
Sol: (1,2,3)	Sol: (5,6,7)	Sol: (1,-1,2)	Sol: (2,1,0)

2. System of Linear Equations

A system of linear equations in n variables (or unknowns) $x_1, x_2, \dots, x_n$ is a set of equations of the form,

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned} \right\} \dots\dots\dots (1)$$

Where the coefficients a_{ij} , $i = 1, 2, \dots, m; j = 1, 2, \dots, n$ of the variables free terms b_i ($i = 1, 2, \dots, m$), are real numbers taken from R. If the b_i are all zeros, then the system (1)

is called homogeneous system of linear equation, If at least one b_i is not zero, then the system (1) is called non-homogeneous system of linear equation.

A sequence of numbers $s_1, s_2, \dots, s_n$ is called solution of the system (1) if $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ is a solution of every equation in the system (1).

Augmented Matrices –

$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$ is the augmented matrix for the system (1).

Row echelon form and Reduced Row echelon form:

A matrix is in **Row Echelon** form if it has the following properties:

- Any row consisting entirely of zeros occurs at the bottom of the matrix.
- For each row that does not contain entirely zeros, the first non-zero entry is 1 (called a leading 1).
- For two successive (non-zero) rows, the leading 1 in the higher row is further left than the leading one in the lower row.

Below is an example of row-echelon form:

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

For **Reduced row echelon** form, the leading 1 of every row contains 0 below and above it's in that column.

An example of reduced row-echelon form:

$$\begin{bmatrix} 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot and Free Variable:

After transforming a system to the row-echelon form, the leading unknowns in the system are called pivot variables, and the other unknowns, are called free variables.

Example 1:

Reduce the following augmented matrix to Reduced Row echelon form (RREF).

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & -5 & 0 & 17 & 29 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \quad (R_2 \sim R_2 - R_3)$$

$$\Rightarrow \begin{bmatrix} 2 & 4 & -5 & 6 & -5 & -1 \\ 0 & 0 & -5 & 0 & 17 & 29 \\ 0 & 0 & -2 & 0 & 7 & 12 \end{bmatrix} \quad (R_1 \leftrightarrow R_3)$$

$$\Rightarrow \begin{bmatrix} 2 & 4 & 0 & 6 & -22 & -30 \\ 0 & 0 & -5 & 0 & 17 & 29 \\ 0 & 0 & -2 & 0 & 7 & 12 \end{bmatrix} \quad (R_1 \sim R_1 - R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & -11 & -15 \\ 0 & 0 & -5 & 0 & 17 & 29 \\ 0 & 0 & -2 & 0 & 7 & 12 \end{bmatrix} \quad (R_1 \sim \frac{R_1}{2})$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & -11 & -15 \\ 0 & 0 & -5 & 0 & 17 & 29 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad (R_3 \sim 5R_3 - 2R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & -11 & -15 \\ 0 & 0 & -5 & 0 & 17 & 29 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad (R_3 \sim 5R_3 - 2R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & -11 & -15 \\ 0 & 0 & -5 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad (R_2 \sim R_2 - 17R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & -11 & -15 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} (R_2 \sim -\frac{1}{5}R_2) \text{ //Row Echelon Form//}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} (R_3 \sim R_1 + 11R_3) \text{ //Reduced Row Echelon Form//}$$

ANS.

Example – 2:

Consider the system

$$2x + 8y + 4z = 2$$

$$2x + 5y + z = 5$$

$$4x + 10y - z = 1$$

Reduce the augmented matrix to RREF.

Solution:

$$\begin{bmatrix} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 4 & 2 & 1 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{bmatrix} (R_1 \sim \frac{R_1}{2})$$

$$\Rightarrow \begin{bmatrix} 1 & 4 & 2 & 1 \\ 2 & 5 & 1 & 5 \\ 0 & 0 & 3 & 9 \end{bmatrix} (R_3 \sim 2R_2 - R_3)$$

$$\Rightarrow \begin{bmatrix} 1 & 4 & 2 & 1 \\ 0 & 3 & 3 & -3 \\ 0 & 0 & 3 & 9 \end{bmatrix} (R_2 \sim 2R_1 - R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} (R_2 \sim \frac{R_2}{3})$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -2 & 5 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} (R_1 \sim R_1 - 4R_2)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 11 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad (R_1 \sim R_1 + 2R_2)$$

$$\begin{bmatrix} 1 & 0 & 0 & 11 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \text{ is the RREF of the given matrix. (ANS.)}$$

Rank of a Matrix:

The rank is how many of the rows are "unique": not made of other rows. (Same for columns.)

Formally, we say, the maximum number of its linearly independent rows (or columns) of a matrix is called the rank of a matrix. The rank of a matrix cannot exceed the number of its rows or columns. To determine the rank of a matrix, we will reduce the given the matrix into row echelon form. Then the number of non-zero rows (or columns) in that matrix will denote the rank of the matrix.

Example – 3 –

Find the rank of the matrix A - $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$.

Solution:

Step 1: Reducing the matrix to row echelon form:

$$\begin{aligned} A &= \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 2 \\ 3 & 0 & 5 \end{bmatrix} [R_2 \sim 2R_1 - R_2] \\ &= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 2 \\ 0 & 6 & 4 \end{bmatrix} [R_3 \sim 3R_1 - R_3] \\ &= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 2 \\ 0 & 0 & 0 \end{bmatrix} [R_3 \sim 2R_2 - R_3] \end{aligned}$$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 0 \end{bmatrix} [R_2 \sim \frac{1}{3}R_2]$$

Which is in Row echelon form. Since there are two non-zero rows, $\text{rank}(A) = 2$. **ANS.**

Example – 4:

Find the rank of the matrix $A = \begin{bmatrix} 6 & 2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{bmatrix}$

Solution:

$$\begin{aligned} & \begin{bmatrix} 6 & 2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 2 & 0 & 4 \\ 0 & -1 & 9 & 16 \\ 0 & -4 & 36 & 64 \end{bmatrix} [R_2 \sim R_1 + 3R_2; \quad R_3 \sim R_1 + 6R_3] \\ &= \begin{bmatrix} 6 & 2 & 0 & 4 \\ 0 & -1 & 9 & 16 \\ 0 & -1 & 9 & 16 \end{bmatrix} [R_3 \sim \frac{R_3}{4}] \\ &= \begin{bmatrix} 6 & 2 & 0 & 4 \\ 0 & -1 & 9 & 16 \\ 0 & 0 & 0 & 0 \end{bmatrix} [R_3 \sim R_2 - R_3] \\ &= \begin{bmatrix} 1 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 1 & -9 & -16 \\ 0 & 0 & 0 & 0 \end{bmatrix} [R_2 \sim -R_2; \quad R_1 \sim \frac{1}{6}R_1] \end{aligned}$$

Which is in Row-echelon form. Since the number of nonzero rows are = 2, $\text{rank}(A) = 2$.
(ANS.)

Exercises:

Transform the following matrices to Row echelon form and then to Reduced Row echelon form –

(1) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & 4 \\ 2 & -4 & 3 \end{bmatrix}$

(2) $\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$

$$(3) \begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -3 \\ 2 & 4 & 4 \end{bmatrix}$$

$$(4) \begin{bmatrix} 1 & 2 & -3 & 1 & 2 \\ 2 & 4 & -4 & 6 & 10 \\ 3 & 6 & -6 & 9 & 13 \end{bmatrix}$$

Determine the rank of the following matrices –

$$(i) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 3 \\ 3 & 6 & 2 & -6 & 5 \end{bmatrix} \quad (iii) \begin{bmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & 2 & 5 \end{bmatrix} \quad (iv) \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$$

$$(vi) \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Solution of Systems of linear equations using Gaussian Elimination Method:

***Reducing augmented matrix to echelon form.**

Solve the following systems of linear equations –

(i)

$$x + y + z = 6$$

$$2x + 3y + 4z = 20$$

$$3x - 2y + z = 2$$

Solution:

The SLE can be written as,

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \\ 2 \end{bmatrix}$$

Augmented Matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 2 & 3 & 4 & 20 \\ 3 & -2 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -5 & -2 & -16 \end{bmatrix} [R_2 \sim R_2 - 2R_1; R_3 \sim R_3 - 3R_1]$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 8 & 24 \end{bmatrix}$$

Now we can write,

$$x + y + z = 6 \Rightarrow x = 1$$

$$y + 2z = 8 \Rightarrow y + 6 = 8 \Rightarrow y = 2$$

$$8z = 24 \Rightarrow z = 3$$

So, the solution is $(x, y, z) = (1, 2, 3)$ **ANS.**

Solution of Systems of linear equations using Gauss-Jordan Elimination Method:

***Reducing augmented matrix to reduced row-echelon form.**

(ii)

$$x + 2y + 3z = 5$$

$$2x - y + z = 5$$

$$4x + 2y - 3z = 5$$

Solution:: The SLE can be written as,

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 4 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$$

Augmented Matrix:

$$\begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & -1 & 1 & 5 \\ 4 & 2 & -3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 5 & 5 & 5 \\ 0 & 6 & 15 & 15 \end{bmatrix} [R_2 \sim 2R_1 - R_2; R_3 \sim 4R_1 - R_3]$$

$$= \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 5 & 5 \end{bmatrix} [R_2 \sim \frac{R_2}{5}, R_3 \sim \frac{R_3}{3}]$$

$$= \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 3 \end{bmatrix} [R_3 \sim R_3 - 2R_1]$$

$$= \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} [R_3 \sim \frac{R_3}{3}]$$

$$= \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} [R_2 \sim R_2 - R_3]$$

$$= \begin{bmatrix} 1 & 0 & 3 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} [R_1 \sim R_1 - 2R_2]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} [R_1 \sim R_1 - 3R_3]$$

Thus we have,

$$x = 2; y = 0, z = 1; \text{ANS.}$$

(iii)

$$x + y + z + t = 2$$

$$2x + 3y + 4z + 5t = 9$$

$$3x - 2y + z + 2t = 9$$

$$x + 2y - 2z + 3t = 5$$

$$\text{Solution: } x = 1, y = -1, z = 0, t = 2$$

(iv)

$$x + y + z - 2t = -4$$

$$x - 2y + 3z + 4t = 10$$

$$2x + 3y - z + 2t = 9$$

$$4x - y + 2z - t = -7$$

$$\text{Solution: } x = -1, y = 2, z = 1, t = 3$$

Determination of Inverse of a Matrix using RREF

(1) Find the inverse of the matrix $A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$ using RREF form.

Solution:

$$\begin{aligned}
 [AI_3] &= \left[\begin{array}{ccc|ccc} -1 & 2 & -3 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 4 & -2 & 5 & 0 & 0 & 1 \end{array} \right] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 3 & -3 & 1 & 1 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 4 & -2 & 5 & 0 & 0 & 1 \end{array} \right] [R_1 = R_1 + R_2] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 3 & -3 & 1 & 1 & 0 \\ 0 & 4 & -5 & 0 & 2 & -1 \\ 0 & 14 & -17 & 4 & 4 & -1 \end{array} \right] [R_2^* = 2R_2 - R_3, R_3^* = 4R_1 - R_3] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 3 & -3 & 1 & 1 & 0 \\ 0 & 4 & -5 & 0 & 2 & -1 \\ 0 & 0 & -\frac{1}{2} & -4 & 3 & -\frac{5}{2} \end{array} \right] [R_3^* = \frac{7}{2}R_2 - R_3] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 3 & -3 & 1 & 1 & 0 \\ 0 & 4 & -5 & 0 & 2 & -1 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right] [R_3^* = -\frac{1}{2}R_3] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 3 & -3 & 1 & 1 & 0 \\ 0 & 4 & 0 & 40 & -28 & 24 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right] [R_2^* = 5R_3 + R_2] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 3 & -3 & 1 & 1 & 0 \\ 0 & 1 & 0 & 10 & -7 & 6 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right] [R_2^* = \frac{1}{4}R_2] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -29 & 22 & -18 \\ 0 & 1 & 0 & 10 & -7 & 6 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right] [R_1^* = R_1 - 3R_2] \\
 &= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -5 & 4 & -3 \\ 0 & 1 & 0 & 10 & -7 & 6 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right] [R_1^* = R_1 + 3R_2]
 \end{aligned}$$

$$\text{So, } A^{-1} = \begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix}$$

(Ans.)

(2) Find the inverse of the following matrix by using reduced row echelon form:

$$A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}$$

(3) If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$, then find the values of $A^{-1}B$ and AB^{-1} .

3. Determinants

Determinants of a square matrix :

A determinant of order n is a square array of n^2 quantities a_{ij} ($i, j = 1, 2, \dots, n$) enclosed between two vertical bars and is generally written in this form given below:

$$D = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{vmatrix} \quad (1)$$

Which is an ordinary number.

Minors and Co-factors:

If in the determinant $D = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{vmatrix}$, we delete the i -th row and j -th column

a determinant from all the elements remaining, we shall have a new determinant of $(n-1)$

rows and $(n-1)$ columns which is called the minor of the element a_{ij} and is denoted by M_{ij} ;

The minor M_{ij} multiplied by $(-1)^{i+j}$ is called the co-factor of a_{ij} in D and will be denoted by $A_{ij} = (-1)^{i+j} M_{ij}$

3.1 Determination of inverse matrix using determinant:

Previously we have seen how to determine the inverse of a matrix using determinants. so, there's no need to discuss about this anymore.

4. Vectors

Definition:

Vectors are mathematical entities, which have both a magnitude and a direction. The location of the vector (for example, on which point a specific vector force is acting, or where a car with a given vector velocity is located) is not part of the vector itself. Physical examples include forces, velocities, momentum and locations. Note that, not all entities having a magnitude and direction are necessarily vectors. For example, the polarization of light and the finite angle rotation etc.

Vector Space:

A vector space over an arbitrary field F is a non-empty set V, whose elements are called vectors for which two operations are prescribed. The first operation, called vector addition, assigns to each pair of vectors u and v in V, a vector denoted by $u + v$, called their sum. The second operation, called scalar multiplication assigns to each vector v in V and scalar α in F a vector denoted by αv which is in V.

For example –

Vector addition –

Let two vectors are $v_1 = (1, 2, -3)$ and $v_2 = (5, -7, 4)$. Sum of the vectors, $V = v_1 + v_2 = (6, -5, 1)$

Scalar Multiplication –

Let $v_1 = (-3, 6, -5)$ multiplied by $a = 7$ gives, $V_2 = (-21, 42, -30)$

4.2 Linear Combination of Vectors

Let V be a vector space over the field F and let $v_1, v_2, \dots, v_n \in V$ then any vector $v \in V$ is called a linear combination of $v_1, v_2, \dots, v_n$ if and only if there exists scalars $\alpha_1, \alpha_2, \dots, \alpha_n$ in F such that $v = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = \sum_{i=1}^n \alpha_i v_i$

Page # 235: Example – 7, 8

4.2.1 Linear Dependence & Independence of Vectors

- ❖ Let V be a vector space over the field F . The vectors $v_1, v_2, \dots, v_n \in V$ are said to be linearly dependent over F if there exists a non-trivial linear combination of them equal to the zero vector 0 . That is $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$ where $\alpha_i \neq 0$ for at least one i .
- ❖ The vectors $v_1, v_2, \dots, v_n \in V$ are said to be linearly independent over F if the only linear combination of them equal to zero vector (0) is the trivial one. That is ,
 $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$ if and only if $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$
of the vectors.

How to determine if a set of vectors is linearly independent.

- Form the matrix whose rows are the given vectors
- Reduce the matrix to row echelon form
- If there is any zero row in the row echelon form then the vectors are linearly dependent.

Page # 259

Examples – 16, 17, 18

Example – 16:

Prove that the set of vectors $\{(2,1,2), (0,1,-1), (4,3,3)\}$ is linearly dependent.

Solution:

Forming the matrix whose rows are the given vectors and then reducing it into row echelon form –

$$\begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & -1 \\ 4 & 3 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} [R_3 \sim R_3 - 2R_1]$$

$$= \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} [R_3 \sim R_3 - R_2]$$

The matrix is in row echelon form and has a zero row, hence the given vectors are linearly dependent.
(Shown)

Example – 17:

Show that the set of vectors $\{(3,0,1,-1), (2,-1,0,1), (1,1,1,-2)\}$ is linearly dependent.

Solution:

Forming the matrix whose rows are the given vectors and then reducing it into row echelon form –

$$\begin{bmatrix} 3 & 0 & 1 & -1 \\ 2 & -1 & 0 & 1 \\ 1 & 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 & -1 \\ 0 & 3 & 2 & -5 \\ 0 & -3 & -2 & 5 \end{bmatrix} [R_2 \sim 2R_1 - 3R_2; R_3 \sim R_1 - 3R_3]$$

$$= \begin{bmatrix} 3 & 0 & 1 & -1 \\ 0 & 3 & 2 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} [R_3 \sim R_1 + R_2]$$

The matrix is in row echelon form and has a zero row, hence the given vectors are linearly dependent.
(Shown)

Example – 18:

Show that the set of vectors $(2,-1,4), (3,6,2), (2,10,-4)$ are linearly independent.

Solution:

Forming the matrix whose rows are the given vectors and then reducing it into row echelon form –

$$\begin{bmatrix} 2 & -1 & 4 \\ 3 & 6 & 2 \\ 2 & 10 & -4 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 4 \\ 0 & -15 & 8 \\ 0 & -11 & 8 \end{bmatrix} [R_2 \sim 3R_1 - 2R_2; R_3 \sim R_1 - R_3]$$

$$= \begin{bmatrix} 2 & -1 & 4 \\ 0 & -15 & 8 \\ 0 & 0 & -32 \end{bmatrix} [R_3 \sim 11R_2 - 15R_3]$$

Since the echelon matrix has no zero row, the vectors are linearly independent.

(Showed)

Exercises – (Page – 271): 1, 2, 3, 6, 15

4.3 Eigenvalues and Eigenvectors

